N MOLECULAR
Experimental Research Portfolio

V3–V11 | Source-derived charts, methods, results, audits and literature context

13,312	paired SNN comparisons audited (V9–V11)
100	archived V6–V8 classification-statistic rows
208	V9–V11 computational condition summaries
81	named V11 tests in preserved historical logs
32.067 s	recorded V11 64-seed pipeline runtime

Research archive snapshot: 23 October 2026

All in-project plots are derived from the recovered files. External benchmarks are shown in a separate literature panel and are not claimed as project outcomes.

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1 Research scope and evidentiary standard

This report consolidates experiments originally recorded from V3 through V11. It describes a compiled native spiking neural network (SNN) receiving graph interventions constructed from molecular sequences, chemical bonds, computed RNA secondary structures, or small-molecule and short RNA-fragment geometries. The central observed quantity in V9–V11 is
emphneural firing disagreement; it is not a direct measurement of drug binding, DNA/RNA folding accuracy, free-energy agreement with experiment, or biological efficacy.

Evidence layers. Raw JSONL execution records and source code are primary computational records; source-derived audit CSVs are secondary reconstructions; test logs and SHA-256 manifests establish reproducibility of particular software outputs. Public PDB entries and thermodynamic references identify scientific input provenance or external validation standards, not proof that the SNN recovered atomic structures.

Non-independence. Network resets under deterministic seeds are computational trials, not biological replicates. A model or graph intervention repeated with changed seeds is not independent molecular evidence. V9, V10 and V11 run distinct interventions and should not be pooled as one biological response distribution.

2 Experiment chronology

Stage	Recorded scope	Interpretation
V3–V5	Six-case native DNA/RNA bridge and controlled runs	Constructed molecular graph experiments
V6	6,144 additional seed trials; 2,048 ring-control trials	Structural graph discrimination, multiple controls
V7–V8	Four added RNA classes; 8,192 additional native trials	Negative and positive readout controls, replays
V9	2,048 paired SMILES graph comparisons	Chemical-bond graph ablation
V10	3,328 paired comparisons	RNA base-pair and 3D conformer contacts
V11	7,936 paired comparisons; 13 MMFF94 optimizations	Selected RNA energy, pseudoknot candidates, post-for

3 Computational methods

3.1 SNN contrast and firing disagreement

The experiment constructs matched native runs with contact edges present and absent. The SNN retains the 118-neuron, 324-synapse original scaffold and advances for 48 ticks in each paired execution. For a 22-target-node bitmask at tick t (intact A_t, intervened B_t), the recorded disagreement fraction is

$$D = \frac{1}{48 \cdot 22} \sum_{t=0}^{47} \mathrm{popcount}(A_t \oplus B_t). \tag{1}$$

The audit recomputes this fraction from both emitted spike-mask arrays and compares it with the recorded fraction. A larger D indicates more changed target-neuron firing positions under the specified intervention, not improved molecular prediction accuracy.

3.2 Graph classifier accuracy and conditional permutation nulls

The pre-V9 six-case study trains binary graph-intervention decoders on one group of deterministic stimulation seeds and evaluates on held-out seeds. For the binary classification label,

$$\text{Accuracy} = \frac{N_{\text{correct}}}{N_{\text{total}}}, \qquad p_{\text{conditional}} = \frac{1 + \#\{a_{\text{null}} \geq a_{\text{obs}}\}}{4097}, \tag{2}$$

using 4,096 sign-swap permutations in the archived conditional null. Holm family-wise corrections were recorded for prespecified comparison families. These p values describe this computational design, not a population of biological molecules.

3.3 RNA structure and molecular mechanics

V10 uses a weighted nested-pair solver for canonical RNA pairing. V11 augments this with selected experimentally sourced nearest-neighbor energy terms and a crossing-stem candidate search; some thermodynamic terms remain simplified. The short RNA-fragment and ligand geometry path runs RDKit MMFF94 minimization and sampled Cartesian finite-difference checks. RNA fragments are four nucleotides long, not complete tertiary RNA folds. The analytical force consistency condition is sampled as

$$\Delta_{\text{grad}} = \max_k \left| \frac{\partial E}{\partial x_k} - \frac{E(\mathbf{x} + h\mathbf{e}_k) - E(\mathbf{x} - h\mathbf{e}_k)}{2h} \right|. \tag{3}$$

The computed energies are force-field/model outputs, not experimental binding free energies.

4 Verified experimental results

4.1 Six-case classification and the 1KF1 ring control

The V6 readouts for the 1KF1 construct reached 196/256 (76.56%) on the fresh-seed cycle-breaking comparison. A ring-preserving comparator reached 116/256 (45.31%), which did not reliably classify the graph intervention. This supports selective sensitivity to the tested cycle organization, not a claim of molecular geometry prediction. The four added V7/V8 RNA cases showed no corrected classification significance.

4.2 Compound and RNA response interpretation

V9 compares intact chemical bond graphs against their computational ablations; V10 uses RDKit-generated spatial contacts; V11 additionally studies post-force contacts and selected pseudoknot structures. Different versions use different interventions. Disagreement values for caffeine or other compounds therefore should not be ranked across versions as improvements in chemical efficacy.

4.3 V11 execution totals

Branch	Paired comparisons	Native resets
Nested RNA	1,280	2,560
RNA crossing-stem candidates	1,280	2,560
Ligand 3D contacts	2,048	4,096
Post-force contacts	3,328	6,656
Total	7,936	15,872

Each reset executes 48 native SNN ticks; therefore the accepted V11 comparison campaign documents 761,856 ticks in 15,872 resets. Its wall time was recorded as 32.067 seconds on the execution host, not as a benchmark against third-party physical simulation programs.

5 Detailed data visualizations

Every exhibit below names the source evidence table and states what the figure does, or does not, measure. Chart images are emitted from the packaged CSVs with documented rendering programs and SHA-256 digests.

5.1 Exhibit 01: Original vs fresh-seed graph classification

Six previously studied DNA/RNA graph interventions; performance is intervention-label classification, not fold accuracy.

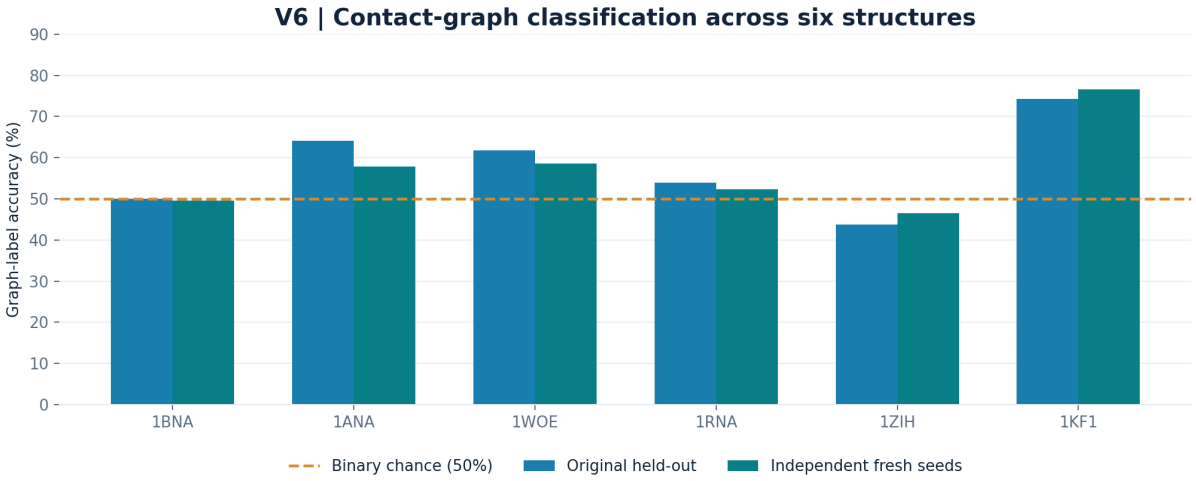


Figure 1: Six previously studied DNA/RNA graph interventions; performance is intervention-label classification, not fold accuracy.

5.2 Exhibit 02: 1KF1 structural ring-control result

The distinct cycle-breaking and cycle-preserving comparisons test sensitivity to the contact-cycle configuration.

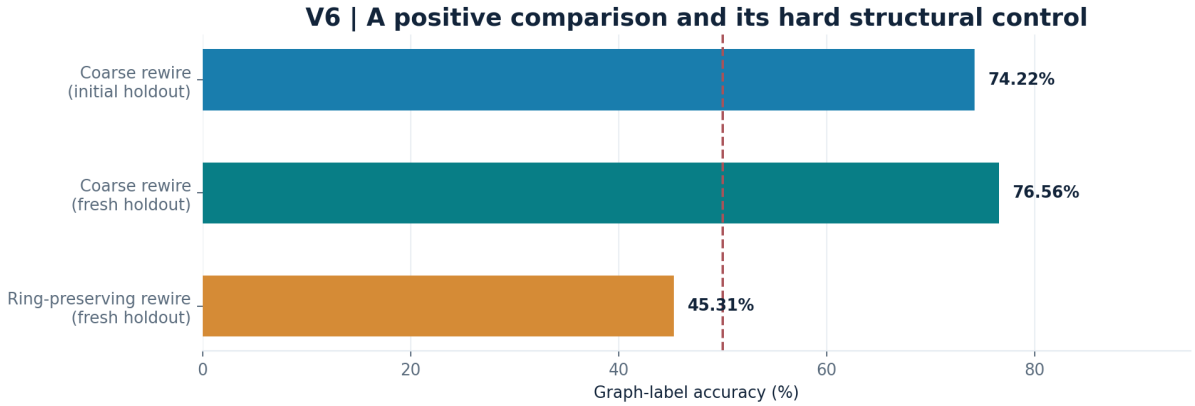


Figure 2: The distinct cycle-breaking and cycle-preserving comparisons test sensitivity to the contact-cycle configuration.

5.3 Exhibit 03: Additional RNA structure labels

Four RNA cases from the V7/V8 expansion, with retained negative results.

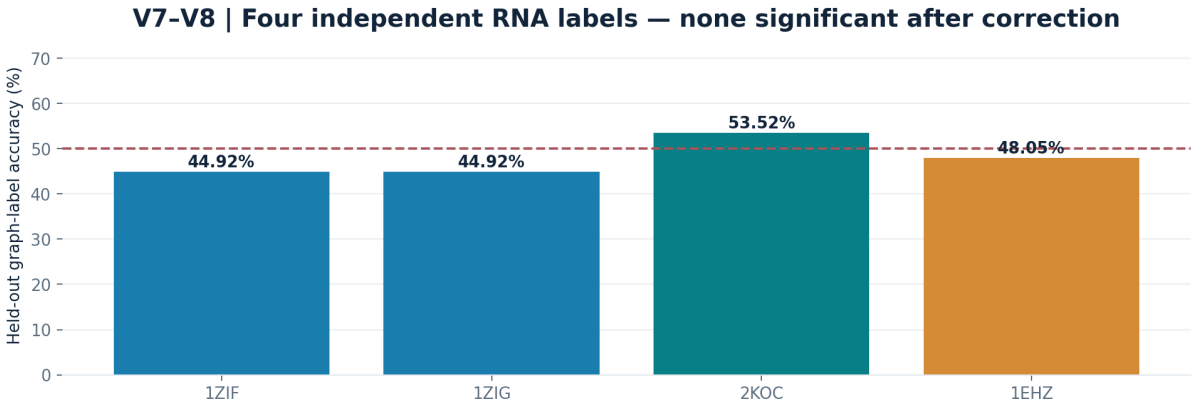


Figure 3: Four RNA cases from the V7/V8 expansion, with retained negative results.

5.4 Exhibit 04: V9 chemical-graph screening

Eight compounds across four feedback and plasticity settings; response is changed target-neuron firing positions.

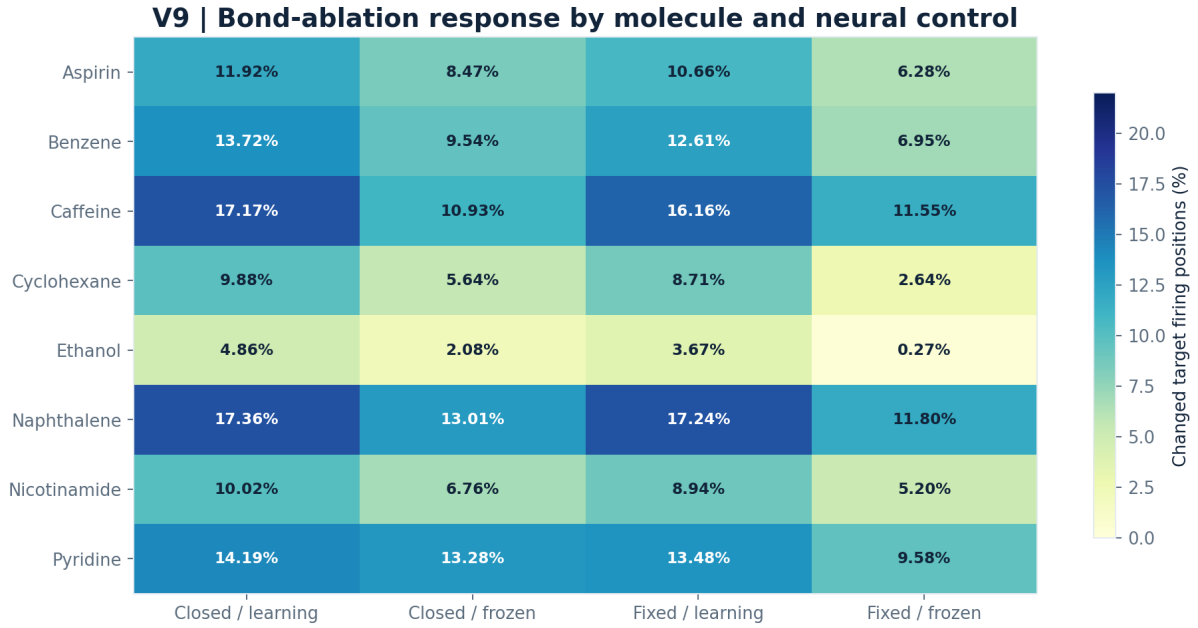


Figure 4: Eight compounds across four feedback and plasticity settings; response is changed target-neuron firing positions.

5.5 Exhibit 05: V10 versus V11 RNA pairing

Algorithmically predicted secondary-structure pair counts; experimental accuracy is not plotted.

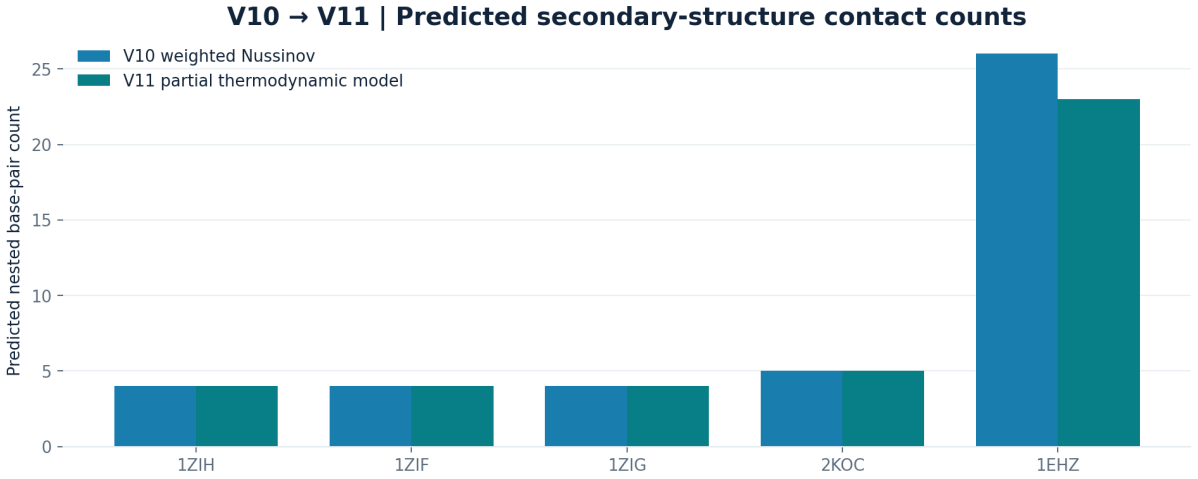


Figure 5: Algorithmically predicted secondary-structure pair counts; experimental accuracy is not plotted.

5.6 Exhibit 06: V11 RNA structural arc visualization

Computed base-pair connectivity rather than a measured molecular structure.

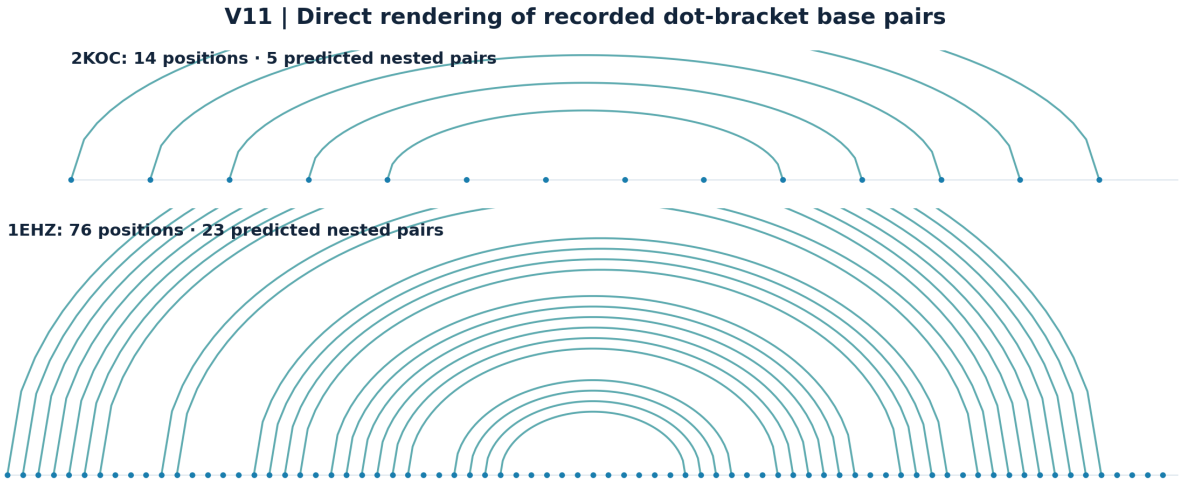


Figure 6: Computed base-pair connectivity rather than a measured molecular structure.

5.7 Exhibit 07: V11 modeled RNA energies

Energy outputs of the implemented partial thermodynamic model, not experimental folding free energies.

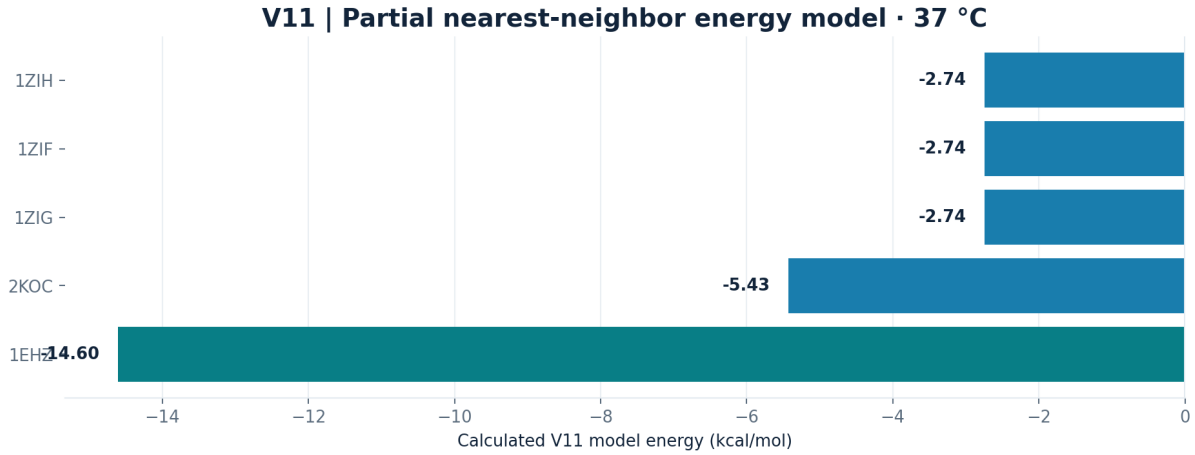


Figure 7: Energy outputs of the implemented partial thermodynamic model, not experimental folding free energies.

5.8 Exhibit 08: Two separate compound response protocols

V9 bond ablation versus V11 spatial-contact ablation; values are not a direct performance comparison.

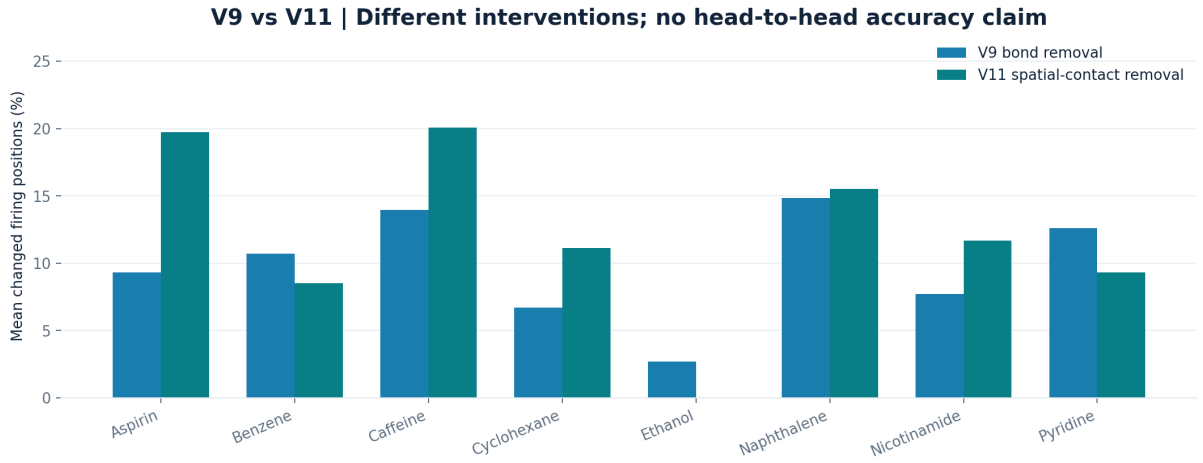


Figure 8: V9 bond ablation versus V11 spatial-contact ablation; values are not a direct performance comparison.

5.9 Exhibit 09: Short RNA-fragment neural response

Four-nucleotide fragment contacts drive the SNN; responses are not full RNA folding accuracy.

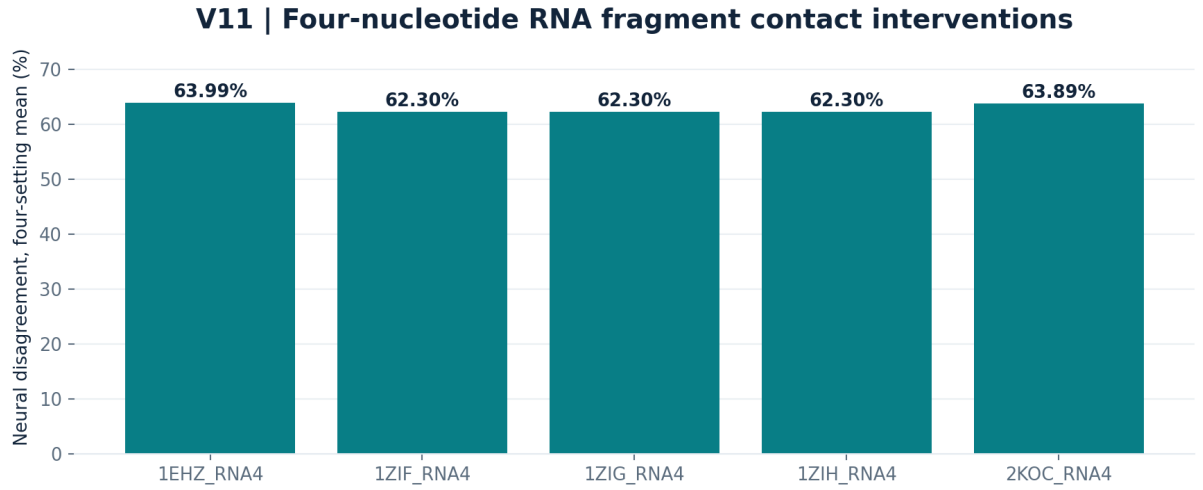


Figure 9: Four-nucleotide fragment contacts drive the SNN; responses are not full RNA folding accuracy.

5.10 Exhibit 10: MMFF94 gradient-consistency checks

Sampled differences between analytical force derivatives and numerical finite differences.

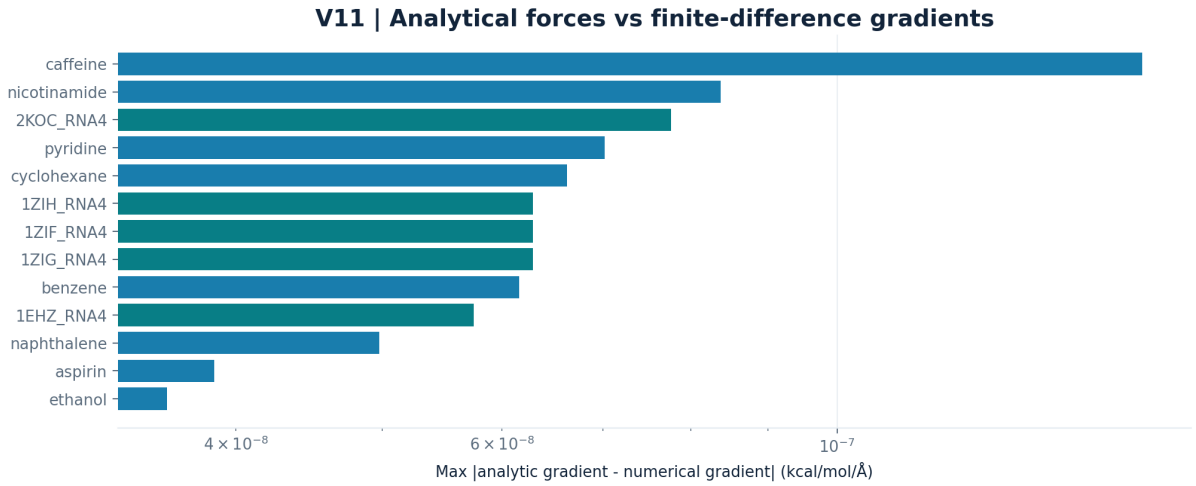


Figure 10: Sampled differences between analytical force derivatives and numerical finite differences.

5.11 Exhibit 11: V11 execution stage timing

Stage-by-stage measured wall times from the accepted 64-seed pipeline report.

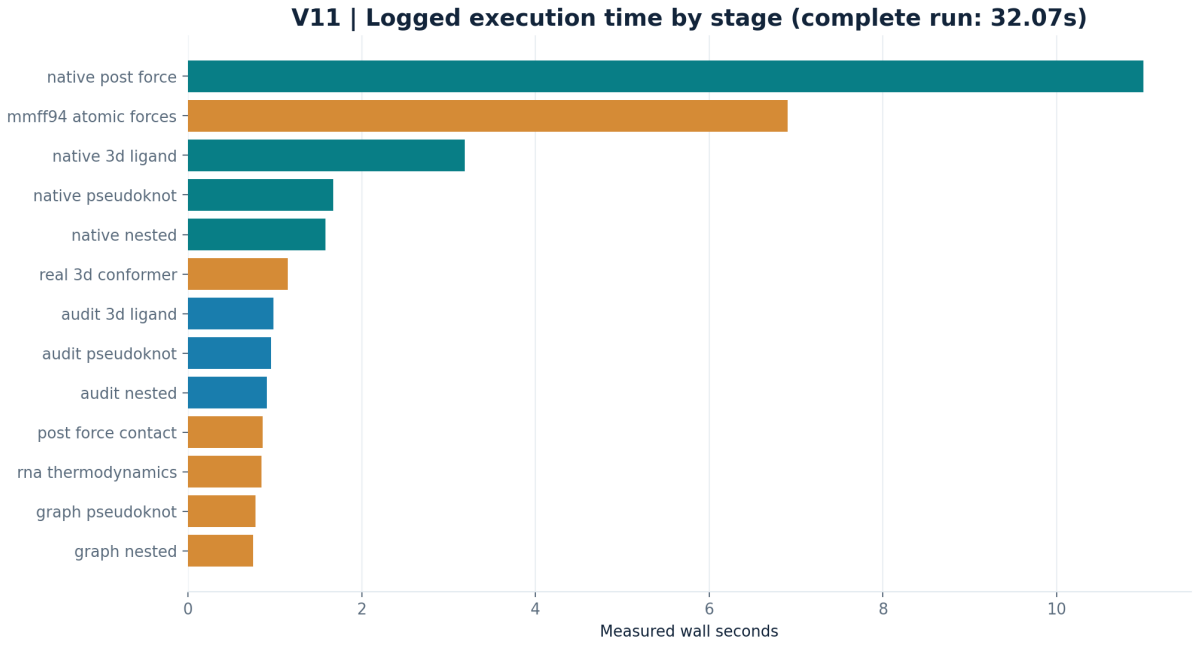


Figure 11: Stage-by-stage measured wall times from the accepted 64-seed pipeline report.

5.12 Exhibit 12: Experimental computation volume

Counts of preserved paired comparisons; no implication of biological sample size.

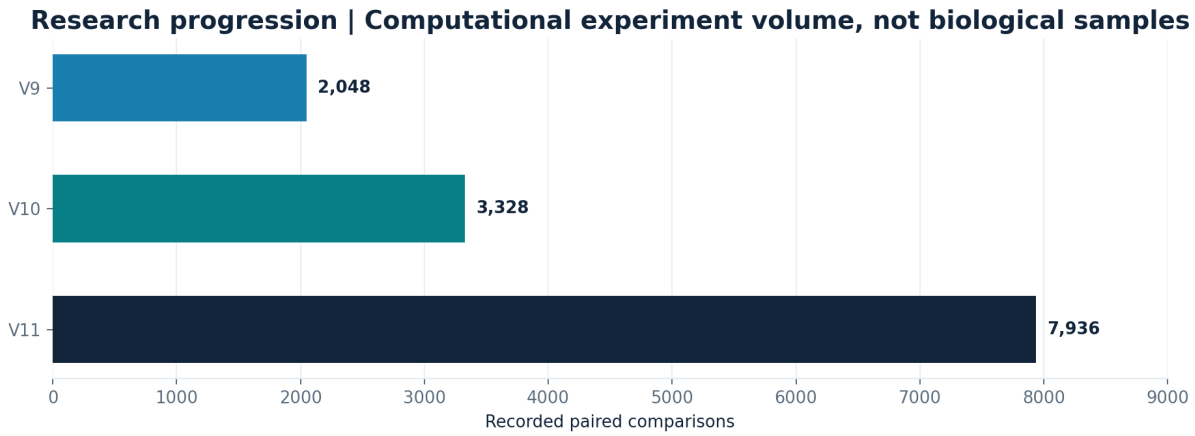


Figure 12: Counts of preserved paired comparisons; no implication of biological sample size.

5.13 Exhibit 13: Recorded regression test suites

Preserved V9, V10 and V11 test logs; not newly executed biological validation.

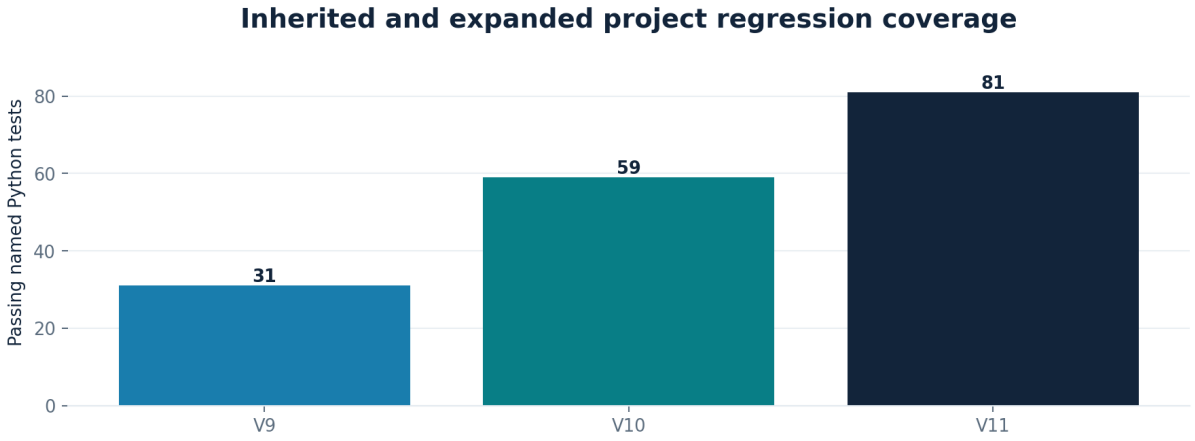


Figure 13: Preserved V9, V10 and V11 test logs; not newly executed biological validation.

5.14 Exhibit 14: All ten reported graph classification cases

Original six and four later cases used different study designs and are explicitly separated.

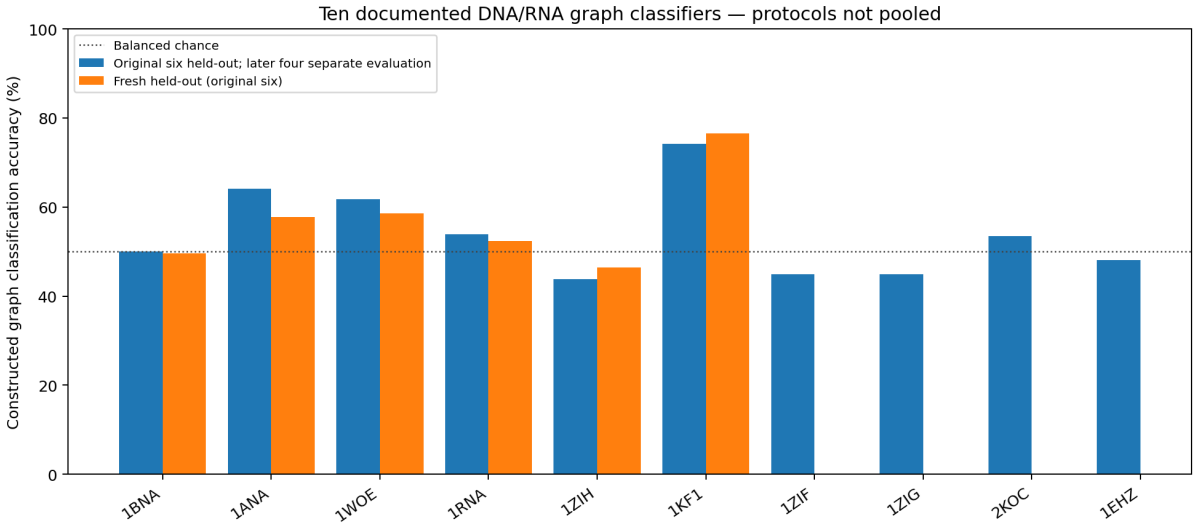


Figure 14: Original six and four later cases used different study designs and are explicitly separated.

5.15 Exhibit 15: V9 complete factorial heatmap

Every compound across fixed/closed feedback and enabled/frozen plasticity.

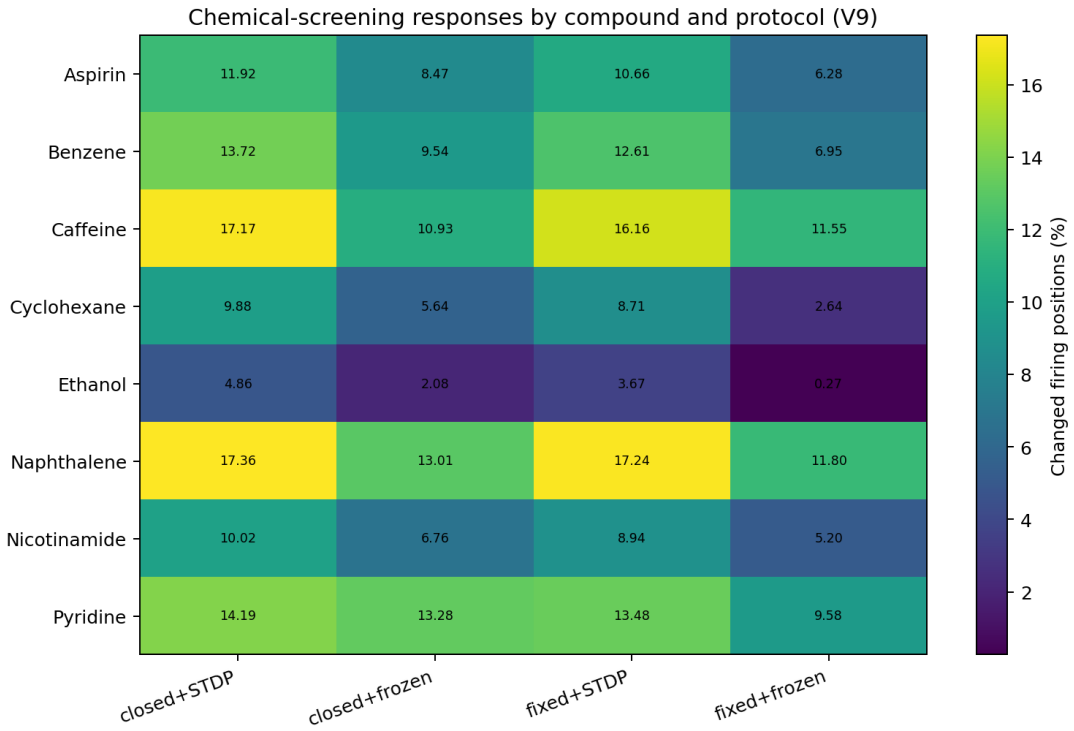


Figure 15: Every compound across fixed/closed feedback and enabled/frozen plasticity.

5.16 Exhibit 16: V10 RNA contact response heatmap

Five sequences and four SNN control settings.

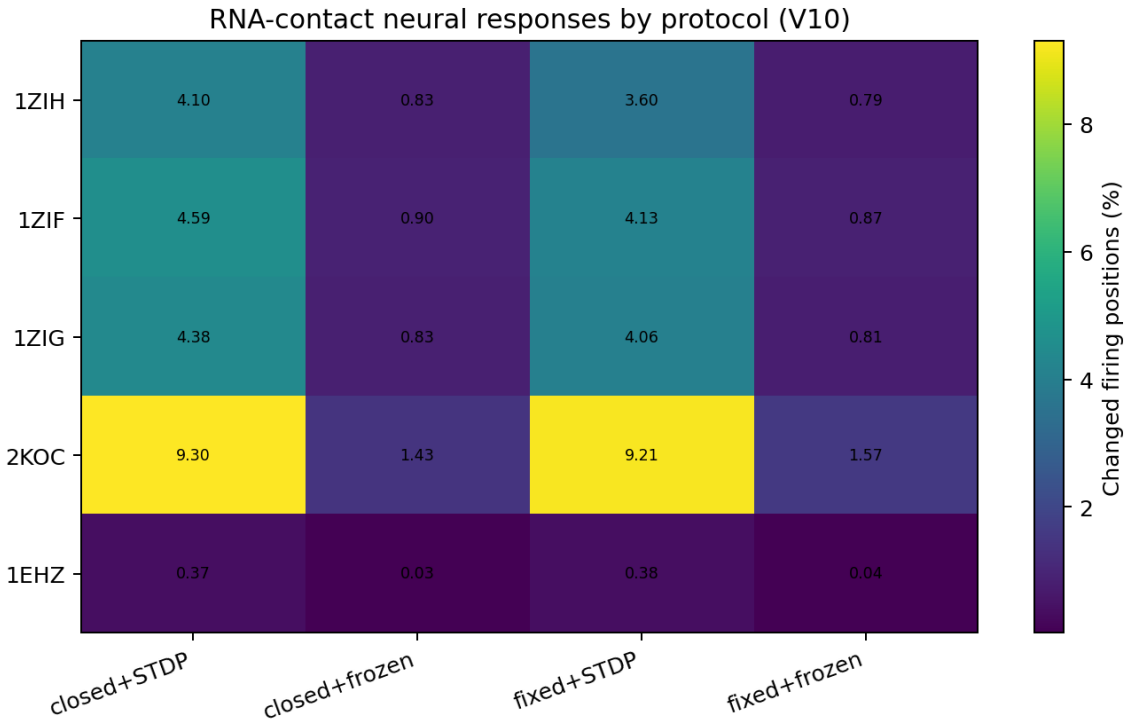


Figure 16: Five sequences and four SNN control settings.

5.17 Exhibit 17: V10 spatial contact counts and response

Descriptive comparison of computed contacts and neural firing changes; not binding prediction.

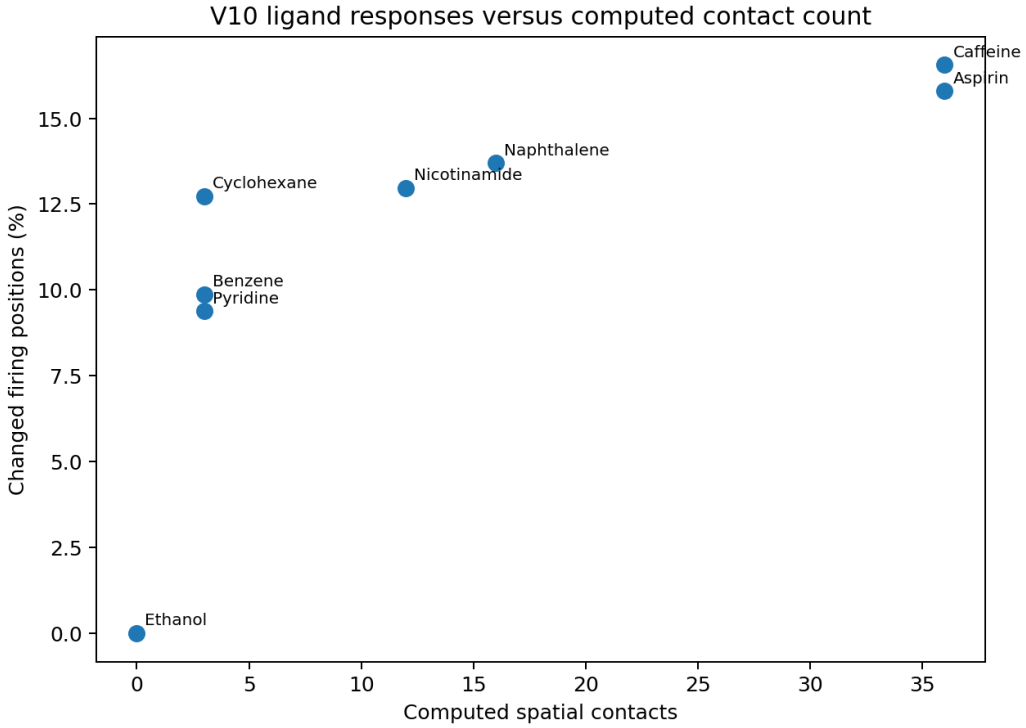


Figure 17: Descriptive comparison of computed contacts and neural firing changes; not binding prediction.

5.18 Exhibit 18: V11 RNA pair counts and model energies

Two distinct computational outputs; no experimental accuracy scores.

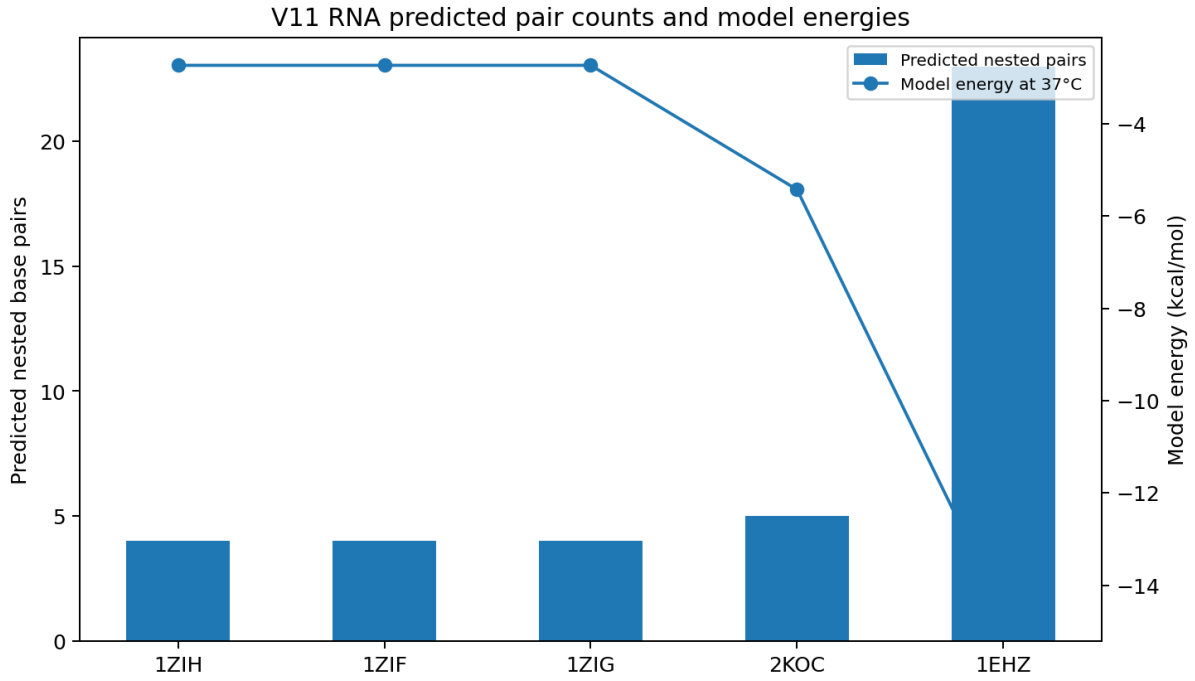


Figure 18: Two distinct computational outputs; no experimental accuracy scores.

5.19 Exhibit 19: V11 post-force contact count

Thirteen RDKit structures, including eight small molecules and five short RNA fragments.

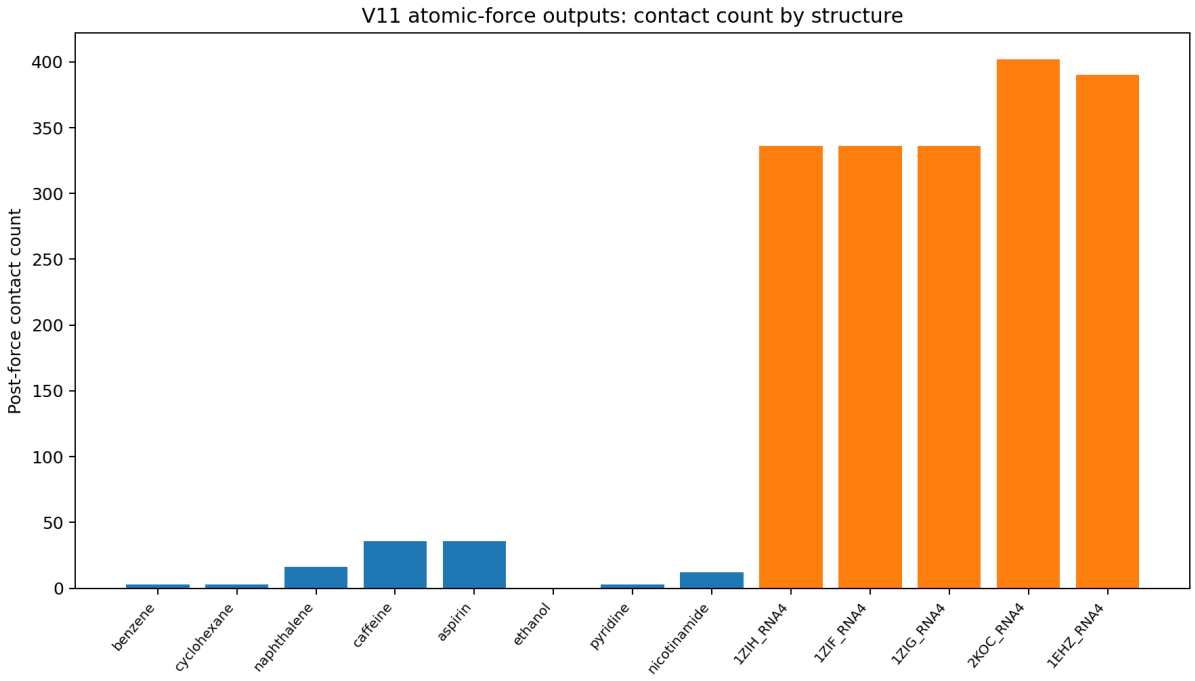


Figure 19: Thirteen RDKit structures, including eight small molecules and five short RNA fragments.

5.20 Exhibit 20: Historical experiment volume and test counts

Computational scale over three releases; overlapping source logic is preserved.

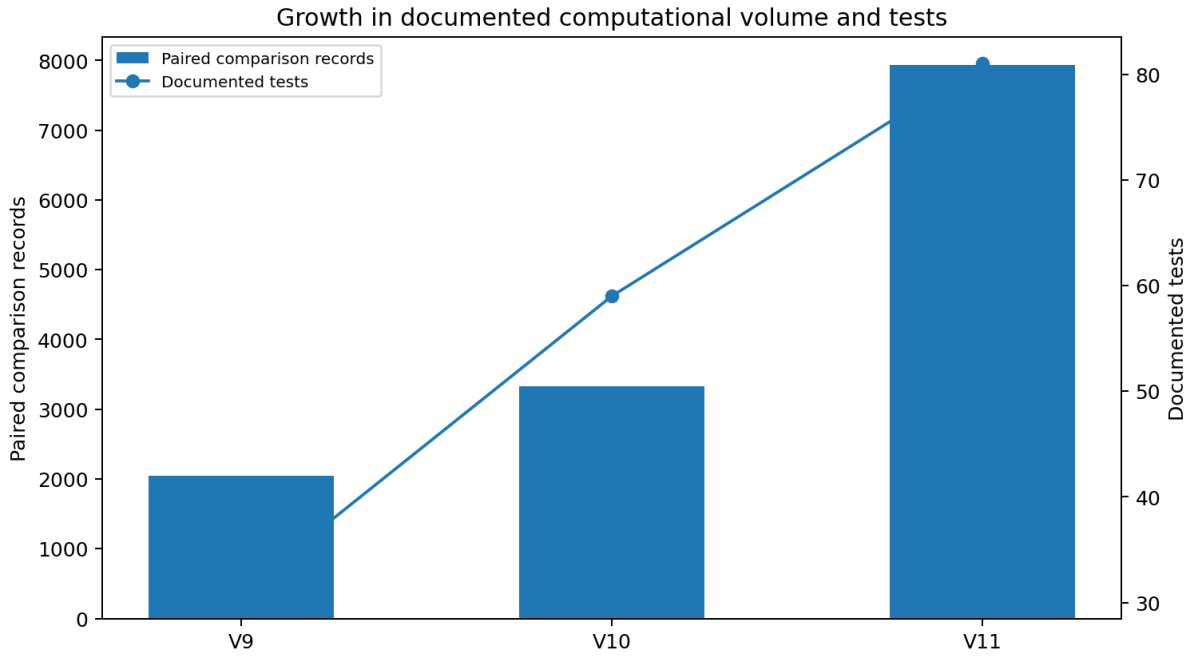


Figure 20: Computational scale over three releases; overlapping source logic is preserved.

5.21 Exhibit 21: External literature reference results

Different published datasets: BPfold on ArchiveII and KnotFold on PKTest; not directly comparable with each other or with the SNN runs.

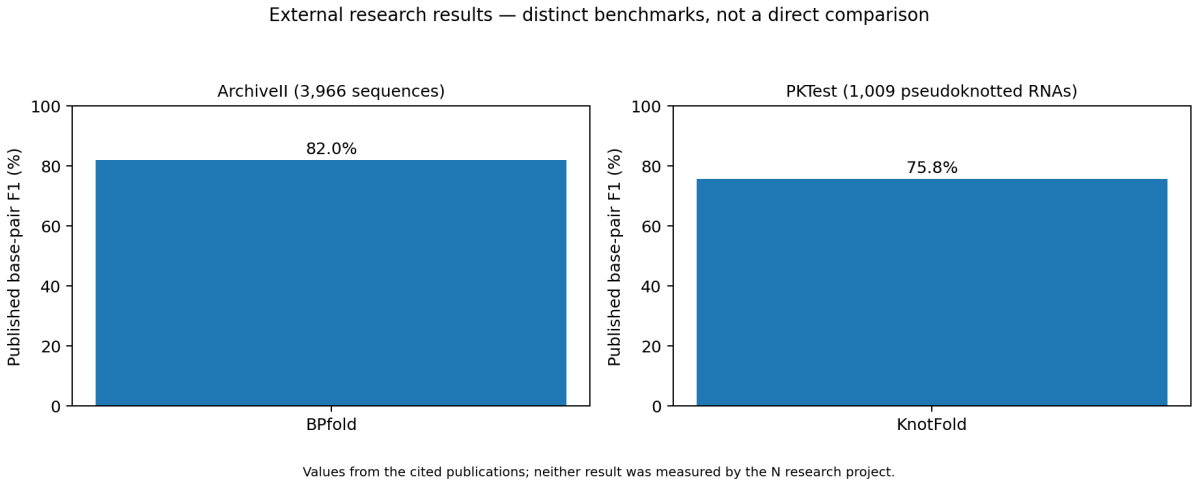


Figure 21: Different published datasets: BPfold on ArchiveII and KnotFold on PKTest; not directly comparable with each other or with the SNN runs.

6 Source-derived numerical summary

6.1 V11 RNA energy outputs

PDB	Length	Pairs	Model energy (kcal/mol)	Crossing candidate
1ZIH	12	4	-2.74000	0
1ZIF	12	4	-2.74000	0
1ZIG	12	4	-2.74000	0
2KOC	14	5	-5.43000	0
1EHZ	76	23	-14.60356	1

Source: `data/RNA_FOLDING_STRUCTURES.csv`. The energy entries do not constitute experimental structure-quality measurements.

6.2 V11 MMFF94 minimized energies and gradients

Structure	Atoms	Final MMFF energy	Max derivative delta	Converged
benzene	12	16.227	6.16e-08	1
cyclohexane	18	-3.561	6.62e-08	1
naphthalene	18	31.133	4.98e-08	1
caffeine	24	-122.528	1.59e-07	1
aspirin	21	18.910	3.87e-08	1
ethanol	9	-1.517	3.60e-08	1
pyridine	11	15.523	7.02e-08	1
nicotinamide	15	5.024	8.37e-08	1
1ZIH_RNA4	135	-463.114	6.29e-08	1
1ZIF_RNA4	135	-463.114	6.29e-08	1
1ZIG_RNA4	135	-463.114	6.29e-08	1
2KOC_RNA4	134	-422.257	7.77e-08	1
1EHZ_RNA4	135	-473.756	5.75e-08	1

Source: `data/ATOMIC_FORCE_RESULTS.csv`. Intermolecular energies are not ligand binding energies.

6.3 V11 timing ledger

Recorded stage	Wall seconds
audit_3d_ligand_seconds	0.9854
audit_nested_seconds	0.9083
audit_pseudoknot_seconds	0.9602
graph_nested_seconds	0.7489
graph_pseudoknot_seconds	0.7813
mmff94_atomic_forces_seconds	6.9057
native_3d_ligand_seconds	3.1853
native_nested_seconds	1.5822
native_post_force_seconds	10.9967
native_pseudoknot_seconds	1.6699

Recorded stage	Wall seconds
post_force_contact_seconds	0.8598
real_3d_conformer_seconds	1.1486
rna_thermodynamics_seconds	0.8488
full_pipeline_wall	32.0671

7 Reproducibility, raw logs and source audit

7.1 Evidence inventory

- `data/ALL_V9_V10_V11_PAIRED_COMPARISONS.csv`: 13,312 computational paired records.
- `data/ALL_CONDITIONS_FROM_RAW.csv`: 208 condition summaries.
- `data/GRAPH_CLASSIFICATION_CONDITIONS.csv`: 100 archived graph-classifier results.
- `data/FIGURE_PROVENANCE.csv`: figure-to-source mapping and prior figure hashes.
- `data/FIGURE_PROVENANCE_EXTENDED.csv`: added figure-to-source mapping.
- `data/FIGURE_SHA256_EXTENDED.csv`: current rendered figure checksums.
- `evidence_and_logs/`: preserved logs, source checks, and a reproducible evidence ZIP.
- `reproduce/`: independent verifier and original chart-rendering scripts.
- `prior_portfolios/`: earlier self-contained portfolio releases retained byte for byte.

7.2 Independent audit outcome

The previous source audit found 0 errors in 13,312 firing-mask disagreement calculations, 208 condition aggregations, and 100 graph-classification-statistic rows. The archived V11 test ledger identifies 81 passing tests, but this portfolio release is a render-and-archive exercise, not a fresh execution of the entire native machine or external molecular benchmarks. The ZIP includes source and data needed for independent checks.

7.3 Scientific interpretation and limitations

The project demonstrates the technical ability to execute native SNN experiments, compute selected RNA secondary-structure representations, generate small-molecule conformers, optimize short fragments, and couple resulting graph contacts into neural execution. It has not established a blind experimental benchmark for folding precision/recall, complete atomistic folding trajectories, measured affinity prediction, or superiority over published structural prediction systems. These distinctions are essential for reproducibility and falsifiability.

8 External scientific literature and data provenance

1. Turner nearest-neighbor energy model (University of Rochester): https://rna.urmc.rochester.edu/NNDB/.

2. RNAstructure thermodynamic tables: https://rna.urmc.rochester.edu/Overview/Parameters.html.

3. ViennaRNA RNAfold MFE and partition-function documentation: https://www.tbi.univie.ac.at/RNA/ViennaRNA/doc/html/man/RNAfold.html.

4. RDKit force-field helpers API: https://www.rdkit.org/docs/source/rdkit.Chem.rdForceFieldHelpers.html.

5. BPfold: "Deep generalizable prediction of RNA secondary structure via base pair motif energy," *Nature Communications* (2025): https://www.nature.com/articles/s41467-025-60048-1.

6. KnotFold: "Accurate prediction of RNA secondary structure including pseudoknots through solving minimum-cost flow with learned potentials," *Communications Biology* (2024): https://www.nature.com/articles/s42003-024-05952-w.

7. G4ShapePredictor: Liew et al., *Scientific Reports* 14, 24238 (2024): https://www.nature.com/articles/s41598-024-74826-2.

8. RCSB experimentally determined RNA structure 1EHZ: https://www.rcsb.org/structure/1EHZ.

The BPfold F1 value of 82.0% is reported on ArchiveII (3,966 RNAs). KnotFold's overall 75.8% F1 is reported on PKTest (1,009 pseudoknotted RNAs). These are independent benchmarks with different datasets. Neither is a score achieved by the project described here.